

4-5.8

4-1 (1) (3) (5) (7) 4-6 (1) (3) (5) (7) 4-8, 10, 13 (4) (7)
4-14, 16, 17, 19, 20 (4) (7), 23, 25, 26, 28, 32

4-5

$$(1) \quad \frac{1}{3} u[n] \xrightarrow{F} \frac{1}{1 - \frac{1}{3} e^{-j\omega}}$$

$$\frac{1}{3} u[n-1] \xrightarrow{F} \frac{e^{-j\omega}}{1 - \frac{1}{3} e^{-j\omega}}$$

$$(3) \quad \begin{aligned} 2^{-n} &\xrightarrow{F} \frac{1}{\frac{1}{2} + j\omega} \\ 2^{-n-1} &\xrightarrow{F} \frac{e^{-j\omega}}{\frac{1}{2} + j\omega} \end{aligned}$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} \left(\frac{1}{2}\right)^n e^{-j\omega n}$$

$$= \frac{1}{1 - \frac{1}{2} e^{-j\omega}} + \frac{e^{-j\omega}}{2 - e^{-j\omega}}$$

$$2^{-n-1} \xrightarrow{F} \frac{2e^{-j\omega}}{2 - e^{-j\omega}} + \frac{1}{2 - e^{-j\omega}}$$

$$(5) \quad \begin{aligned} \delta[n] &\xrightarrow{F} 1 \\ \delta[n-2] + \delta[n+2] &\xrightarrow{F} e^{-j2\omega} + e^{j2\omega} = 2\cos 2\omega \end{aligned}$$

$$(7) \quad \begin{aligned} a^n u[n] &\xrightarrow{F} \frac{1}{1 - ae^{-j\omega}} \\ a^n \cos \omega_0 n u[n] &\xrightarrow{F} \frac{1}{2} \left[\frac{1}{1 - ae^{-j(\omega - \omega_0)}} + \frac{1}{1 - ae^{-j(\omega + \omega_0)}} \right] \end{aligned}$$

4-6

$$(1) \quad X[n] = \frac{\sin \omega_c n}{\pi n}$$

$$(2) \quad X[n] = n(-1)^n (u[n] - u[n-3])$$

$$(3) \quad \begin{aligned} \frac{\sin \pi n}{\pi n} &\xrightarrow{F} \frac{1}{\pi} \int_{-\pi}^{\pi} e^{j\omega n} d\omega \\ X[n] = \frac{\sin \pi(n - \frac{1}{2})}{\pi(n - \frac{1}{2})} &\xrightarrow{F} X(e^{j\omega}) \end{aligned}$$

(4)

$$\begin{aligned} X[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \sum_{k=0}^3 (-1)^k e^{jk\frac{\pi}{2}n} \end{aligned}$$

(7)

$$X(e^{j\omega}) = \sum_{n=0}^7 \frac{1}{2^n} e^{-j\omega n}$$

$$X[n] = \begin{cases} \frac{1}{2^n} & 0 \leq n \leq 7 \\ 0 & \text{elsewhere} \end{cases}$$

4-8

(1)

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{+\infty} X[n] = -1 + 1 + 2 + 1 + 1 + 1 + 2 + 1 \\ &= 5 + 1 \\ &= 6 \end{aligned}$$

(2)

$$X[n+2] \xrightarrow{F} X(e^{j\omega}) e^{j2\omega} \text{ 无虚部}$$

$$X(e^{j\omega}) = |X(e^{j\omega})| e^{j\theta(\omega)}$$

$$\theta(\omega) + 2\omega = 0$$

$$\theta(\omega) = -2\omega \quad -\pi < \omega < \pi \text{ 其余区间周期平移}$$

(3)

$$X[0] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) d\omega = 2$$

$$\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega = 4\pi$$

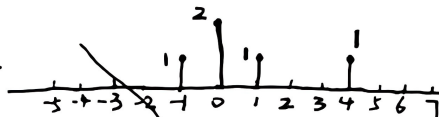
$$(4) \quad X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} X[n] (-1)^n$$

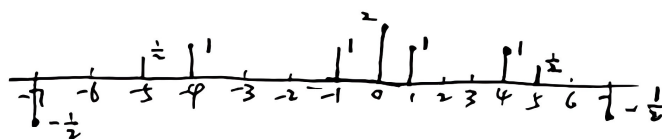
$$= 1 - 1 + 2 - 1 = 1 + 2 - 1 + 1$$

$$= 1 - 1 + 2$$

$$= 2$$

(5)

$$\frac{X[n] + X[-n]}{2}$$




(6)

$$\int_{-\pi}^{\pi} |x(e^{j\omega})|^2 d\omega = 2\pi \sum_{n=-\infty}^{\infty} |x[n]|^2$$

$$= 2\pi (1+1+4+1+1+4+1)$$

$$= 28\pi$$

(7)

$$x[n] \xrightarrow{F} j \frac{dx(e^{j\omega})}{d\omega}$$

$$\int_{-\pi}^{\pi} \left| \frac{dx(e^{j\omega})}{d\omega} \right|^2 d\omega = 2\pi \sum_{n=-\infty}^{\infty} n^2 |x[n]|^2$$

$$= 2\pi (1+1+1+9+32+25+49)$$

$$= 5(2\pi)$$

4-13

(2)

$$x[n] \xrightarrow{F} x(e^{j\omega})$$

$$x[n] \cos \omega_0 n \xrightarrow{F} \frac{1}{2} \{ x(e^{j(\omega-\omega_0)}) + x(e^{j(\omega+\omega_0)}) \}$$

(3)

$$x^*[n] \xrightarrow{F} x^*(e^{j\omega})$$

$$x^*[n] \xrightarrow{F} x^*(e^{j\omega})$$

$$x_3[n] \xrightarrow{F} \frac{x(e^{j\omega}) + x^*(e^{j\omega})}{2} = \text{Re}\{x(e^{j\omega})\}$$

4-14

$$x(e^{j\omega}) \quad x^*(e^{-j\omega}) \quad x(e^{-j\omega}) \quad x[n]$$

$$(1) \sum_{k=1}^{\infty} \sin k\omega$$

$$x(e^{j\omega}) = e^{-j\omega} \sum_{k=1}^{\infty} \sin k\omega$$

$$x^*(e^{-j\omega}) = -e^{-j\omega} \sum_{k=1}^{\infty} \sin k\omega$$

$$x(e^{-j\omega}) = -e^{-j\omega} \sum_{k=1}^{\infty} \sin k\omega$$

$$x[n] \text{ 纯虚, 非奇非偶}$$

$$(2) j \sin n \cos n \omega$$

$$j \sin n \omega \cos n \omega$$

$$-j \sin n \cos n \omega$$

$$\text{实奇函数}$$

$$(3) A e^{jB\omega}$$

$$A e^{jB\omega}$$

$$A e^{jB\omega}$$

$$\text{实, 非奇非偶}$$

$$A e^{jB\omega} e^{jB\omega} = -e^{j\frac{3}{2}\omega} \quad 0 \leq \omega \leq \frac{\pi}{2}$$

4-16

① 要求 $x[n]$ 为奇信号, (5) (7) ✓② 要求 $x[n]$ 为偶信号, (2) (3) (6) ✓③ $x[1] = 0$ (5)④ $\sum_{n=-\infty}^{\infty} x[n] \neq 0$ (1) (4) (3) (4) (6) ✓⑤ $x(e^{j\omega}) = (x(e^{-j\omega}))^*$ $x[n]$ 为实信号

$$\theta(\omega) - \theta(-\omega) = 2\theta(\omega) \Rightarrow \theta(\omega) = -\theta(-\omega)$$

 $\theta(\omega)$ 是 ω 的奇函数 $x[n]$ 经过移位后是偶函数

(2) (3) (4) (6) (7)

4-10

(1)

$$\frac{\sin \frac{\pi n}{3}}{\pi n} \xrightarrow{F} \text{图}$$

$$\frac{\sin \frac{\pi n}{4}}{\pi n} \xrightarrow{F} \text{图}$$

$$\frac{\sin \frac{\pi n}{5}}{\pi n} \cdot \frac{\sin \frac{\pi n}{4}}{\pi n} \xrightarrow{F} \frac{1}{2\pi} \int_{-\pi}^{\pi} x(e^{j\omega}) x^*(e^{j\omega}) d\omega$$

$$= \text{图}$$

(2)

$$a^* u[n] \xrightarrow{F} \frac{1}{1 - a e^{-j\omega}}$$

$$n a^* u[n] \xrightarrow{F} j \frac{a e^{-j\omega}}{(1 - a e^{-j\omega})^2}$$

$$\text{原} (n+1) a^* u[n] \xrightarrow{F} \frac{1}{1 - a e^{-j\omega}} + \frac{a e^{-j\omega}}{(1 - a e^{-j\omega})^2}$$

$$= \frac{1}{(1 - a e^{-j\omega})^2}$$

(3)

$$x[n] = \text{图} * \text{图}$$

$$x(e^{j\omega}) = \left(\frac{E}{M} \right) \left(\frac{\sin \frac{N\omega}{2}}{\sin \frac{\omega}{2}} \right)^2$$

4-17

$$1 \rightarrow \sum_{k=-\infty}^{+\infty} \delta(\omega - 2k\pi)$$

$$\frac{3}{2} \rightarrow \sum_{k=-\infty}^{+\infty} 3\pi \delta(\omega - 2k\pi)$$



$$y[n] \xrightarrow{F} \frac{\sin \frac{3}{2}\omega}{\sin \frac{\omega}{2}}$$

$$\sum_{k=-\infty}^n y[k] \xrightarrow{F} \frac{\sin \frac{3}{2}\omega}{\sin \frac{\omega}{2}} + 3\pi \sum_{k=-\infty}^{+\infty} \delta(\omega - 2k\pi)$$

$$x[n] = \sum_{k=-\infty}^n y[k] = \delta[n+1] + 2\delta[n] + 3u[n-1]$$

4-19

(1)

$$\frac{\sin \frac{\pi}{4}n}{\pi n} \xrightarrow{F} \text{rect}_{\frac{\pi}{4}}$$

$$\left(\frac{\sin \frac{\pi}{4}n}{\pi n}\right)^2 \xrightarrow{F} \text{tri}_{\frac{\pi}{2}}$$

$$\frac{\pi}{2} \leq |W_c| \leq \pi$$

(2)

$$\frac{\sin \frac{\pi}{4}n}{\pi n} \cos \frac{\pi}{2}n \xrightarrow{F} \text{rect}_{\frac{\pi}{4}} \cos \frac{\pi}{2}n$$

$$\frac{3}{4}\pi \leq |W_c| \leq \pi$$

$$4-20 \quad (b) \quad x_1(e^{j\omega}) = \frac{1}{2} [x(e^{j\omega-\pi}) + x(e^{j\omega+\pi})]$$

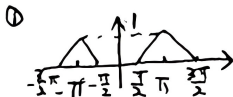
$$x_1[n] = x[n] \cos \pi n$$

$$(f) \quad x_5(e^{j\omega}) = x_1(e^{j\omega}) + x(e^{j\omega})$$

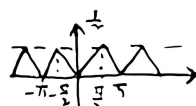
$$x_5[n] = x[n] (\cos \pi n + 1)$$

4-23

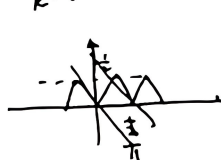
(1)



(2)



$$\sum_{k=-\infty}^n \delta[n-2k] \xrightarrow{F} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\pi) = \sum_{k=-\infty}^{+\infty} \frac{1}{\pi} e^{jk\pi}$$

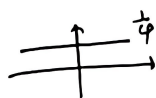


(3)

$$\sum_{k=-\infty}^n \delta[n-4k] \xrightarrow{F} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\frac{\pi}{2}) = \frac{1}{4} \sum_{k=-\infty}^{+\infty} e^{jk\frac{\pi}{2}}$$

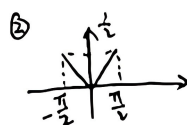
$$W(e^{j\omega}) = \frac{1}{4} \left\{ x(e^{j\omega}) + x(e^{j(\omega-\frac{\pi}{2})}) + x(e^{j(\omega+\frac{\pi}{2})}) \right\}$$

$$= \frac{1}{4}$$

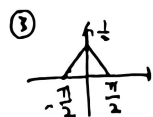


(4)

$$\textcircled{1} \quad y[n] = 0$$



$$y[n] = \frac{\sin \frac{\pi}{2}n}{2\pi n} - 2 \left(\frac{\sin \frac{\pi}{4}n}{\pi n} \right)^2$$



$$y[n] = 2 \left(\frac{\sin \frac{\pi}{4}n}{\pi n} \right)^2$$



$$y[n] = \frac{\sin \frac{\pi}{2}n}{4\pi n}$$

4-25

$$y[n] + \frac{1}{2}y[n-1] - \frac{1}{6}y[n-2] = x[n] - x[n-1]$$

$$(1) \quad Y(e^{j\omega}) + \frac{1}{2}e^{-j\omega}Y(e^{j\omega}) - \frac{1}{6}e^{-2j\omega}Y(e^{j\omega}) = X(e^{j\omega}) - e^{-j\omega}X(e^{j\omega})$$

$$= X(e^{j\omega}) - e^{-j\omega}X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1 - e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega} - \frac{1}{6}e^{-2j\omega}}$$

$$(2) \quad H(e^{j\omega}) = -\frac{2}{5} \frac{1}{1} \left(\frac{-2}{3 - e^{-j\omega}} + \frac{3}{2 + e^{-j\omega}} \right)$$

$$h[n] = \frac{6}{5} \left[-\frac{2}{3} \left(\frac{1}{3} \right)^n u[n] + \frac{3}{2} \left(-\frac{1}{2} \right)^n u[n] \right]$$

$$= \left[-\frac{4}{5} \left(\frac{1}{3} \right)^n + \frac{9}{5} \left(-\frac{1}{2} \right)^n \right] u[n]$$

$$(3) \quad Y(e^{j\omega}) = 6 \frac{1}{1 - \frac{1}{4}e^{-j\omega}} \frac{1 - e^{-j\omega}}{(3 - e^{-j\omega})(2 + e^{-j\omega})}$$

$$= \frac{6}{1 + \frac{1}{2}e^{-j\omega}} + \frac{3}{1 - \frac{1}{4}e^{-j\omega}} + \frac{-16}{5} \frac{1}{1 - \frac{1}{3}e^{-j\omega}}$$

$$y[n] = \left[\frac{6}{5} \left(\frac{1}{2} \right)^n + 3 \left(\frac{1}{4} \right)^n - \frac{16}{5} \left(\frac{1}{3} \right)^n \right] u[n]$$

4-26

$$(1) \quad X(e^{j\omega}) = \frac{1}{1 - \frac{2}{3}e^{-j\omega}}$$

$$Y(e^{j\omega}) = j \frac{dX(e^{j\omega})}{d\omega} = \frac{2}{3} \frac{e^{-j\omega}}{(1 - \frac{2}{3}e^{-j\omega})^2}$$

$$H(e^{j\omega}) = \frac{2}{3} \frac{e^{-j\omega}}{1 - \frac{2}{3}e^{-j\omega}} = \frac{2e^{-j\omega}}{3 - 2e^{-j\omega}}$$

$$(2) \quad 3y[n] - 2y[n-1] = 2x[n-1]$$

4-28

$$y[n] = x[n] * h_1[n] * h_2[n] + (x[n] - x[n-1]) * h_1[n] * h_3[n]$$

$$h[n] = h_1[n] + h_1[n] + (x[n] - x[n-1]) * h_3[n]$$

$$= h_1[n] + h_1[n] + (h_1[n] - h_1[n-1])$$

$$(1) \quad X(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - k\pi)$$

$$Y(e^{j\omega}) = 0 \quad y[n] = 0$$

$$(2) \quad \frac{\pi}{3} < \omega < \frac{5\pi}{12} \text{ 只有 } \frac{3\pi}{8}$$

$$y[n] = \sin\left(\frac{3\pi}{8}n + \frac{\pi}{4}\right)$$

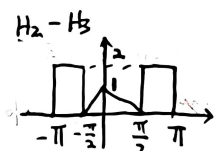
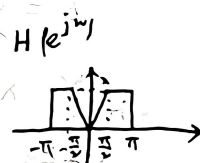
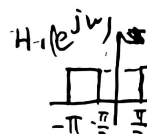
$$(3) \quad \pm 0, \pm \frac{2\pi}{3}, \pm \frac{4\pi}{3}, \pm \frac{5\pi}{3}, \pm \frac{7\pi}{3}$$

$$\text{都不在 } \frac{\pi}{3} \leq \omega \leq \frac{5\pi}{12} \text{ 范围}$$

$$y[n] = 0$$

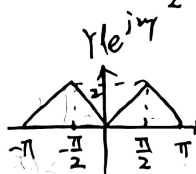
4-32

$$h[n] = h_1[n] * (h_2[n] - h_3[n]) + h_3[n]$$



$$H(e^{j\omega}) = \begin{cases} 2 & \frac{\pi}{2} \leq \omega \leq \pi \\ \frac{4}{\pi}|\omega| & |\omega| < \frac{\pi}{2} \end{cases}$$

$$(-\pi \leq \omega \leq \pi)$$



$$y[n] = \frac{2}{\pi^2} \left(\frac{\pi \sin \frac{\pi}{2} n}{n} - \frac{2}{n^2} \right) [1 + (-1)^n]$$